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Breaking the Black Box: Inherently Interpretable Physics-Informed Machine Learning for Imbalanced Seismic Data

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Abstract

Earthquakes are the most damaging natural disasters in the past century, making ground motion models (GMMs) critical components in seismic risk mitigation and infrastructure design. With rapidly expanding strong motion databases, machine learning (ML) models are increasingly applied to GMM development. However, existing ML-based GMMs rely on post-hoc explanations (e.g., SHAP) to understand model behavior, creating opacity that undermines confidence in high-stakes engineering decisions. Moreover, seismic datasets exhibit severe imbalance, with scarce large-magnitude near-field records compared to abundant small-magnitude far-field data, causing systematic underprediction of critical damage-inducing scenarios. Despite these fundamental limitations, research addressing both interpretability and data imbalance remains limited. This study develops an inherently interpretable neural network architecture employing independent additive pathways coupled with a novel HazBinLoss function. HazBinLoss integrates physics-informed weighting with inverse bin count scaling to address underfitting in sparse, high-hazard regions systematically. The developed model achieves robust performance metrics: mean squared error = 0.5402, mean absolute error = 0.5795, and coefficient of determination = 88.51%. Pathway scaling analysis confirms adherence to established seismological principles across all spectral periods. Comparative SHAP analysis reveals significant limitations, with systematic underestimation of multiple pathway contributions, validating the necessity of inherently interpretable architectures. Nonlinear mixed-effects analysis demonstrates unbiased residuals with sigma components ranging from 0.4033–0.5463 (inter-event), 0.1641–0.4467 (inter-region), 0.5542–0.6892 (intra-region-event), and 0.7449–0.9460 (total). Model predictions exhibit strong agreement with NGA-West2 GMMs and observed data across diverse input conditions. This interpretable framework establishes a foundation for transparent, physics-consistent ML adoption in seismic hazard assessment and risk-informed decision making.

Plain Language Summary

Ground motion models (GMMs) predict how strongly the ground will shake during an earthquake. They are essential for structural analysis, seismic design, and seismic risk assessment studies. Traditional machine learning (ML) approaches are popular to develop GMMs, due to large earthquake databases worldwide. However, they operate as “black boxes,” which are hard to interpret and trust, limiting their use in high-stake decisions. Additionally, these databases suffer from significant data imbalances: fewer large, critically damaging records near the fault compared to abundant, less severely damaging distant records. These two limitations are addressed in this work by developing a transparent ML architecture using the HazBinLoss function. Each input (e.g., magnitude, distance, their interaction term, etc.) is processed separately and added linearly to obtain the output, resulting in exact contribution of each term. The HazBinLoss function assigns higher weights to critical near-field large magnitude records and lower weights to less-critical far-field smaller magnitude records, during training to prevent underprediction of the most damaging scenarios. Our model captures known seismological principles and achieves comparable performance with established GMMs while maintaining transparency. This framework enables broader adoption of ML-based approaches for risk assessment studies and disaster planning.

Keywords: Interpretable Network; Seismic Data Imbalance; Ground Motion Model; Machine Learning; HazBinLoss; Physics-Informed Neural Networks

Highlights

- Developed interpretable ML model for seismic response using independent additive pathways that capture scaling and attenuation
- Introduced HazBinLoss function with physics-informed weighting to avoid underpredicting critical high-magnitude scenarios

- Inherent interpretability outperforms post-hoc SHAP explanations which systematically underestimate component contributions

1 Introduction

Earthquakes are the most damaging natural disasters, accounting for 25.6% of global economic disaster losses since 1900 (United Nations Office for Disaster Risk Reduction, 2025). This seismic risk is mitigated through probabilistic seismic hazard analysis (PSHA), which relies on ground motion models (GMMs) to develop predictive equations for key intensity measures such as the spectral acceleration (PSA) and their associated uncertainties (Baker et al., 2021). Traditionally, GMMs are developed empirically by assuming a functional form based on underlying seismological principles that incorporates source parameters (e.g., moment magnitude M_w , fault type), path effects (e.g., distance terms), and site conditions (e.g., shear-wave velocity V_{s30}) to estimate these measures, including single-station sigma and total sigma for aleatory uncertainty quantification (Al Atik et al., 2010; Baker et al., 2021).

Prominent traditional GMMs, such as those by ASK14: Abrahamson et al. (2014), BSSA14: Boore et al. (2014), CB14: Campbell and Bozorgnia (2014), and CY14: Chiou and Youngs (2014), were developed using the global shallow crustal NGA-West2 dataset. Lack of universal strict guidelines for selecting the number of input parameters leads to variability: simpler models like BSSA14 incorporate limited terms, such as magnitude scaling and style of faulting for the source term, regionalized geometric spreading and anelastic attenuation for path effects, basin effects, and regionalized linear and nonlinear site response. In contrast, more complex models like ASK14 additionally consider hanging-wall effects, depth to top of rupture (Z_{TOR}), directivity effects (through not explicitly modelled), and aftershock scaling, which can significantly complicate the GMM and potentially increase overfitting risks (Bindi, 2017). In summary, these traditional approaches employ diverse parameterizations to predict intensity measures and quantify associated uncertainties; however, fixed functional forms can limit flexibility in capturing complex data patterns.

Addressing these limitations, with the large volume of records in recent ground motion databases, alternative data-driven machine learning (ML) approaches have emerged for GMM development (e.g., Sreenath et al., 2023; Meenakshi et al., 2023; Mohammadi et al., 2023; Sedaghati and Pezeshk, 2023; Zhu et al., 2023; Fayaz et al., 2024; Somala et al., 2024; Sreenath et al., 2024; Alidadi and Pezeshk, 2025; Ding et al., 2025). As nonparametric methods, these approaches impose fewer assumptions on the data, learn highly nonlinear patterns, and perform well with larger datasets, potentially resulting in improved generalization over traditional GMMs. ML approaches excel in interpolation but perform poorly in extrapolating unseen magnitude and distance values, particularly without additional constraints. One effective method to ensure the targets satisfy appropriate scaling and attenuation constraints on magnitude, depth, distance, and V_{s30} is to implement monotonic constraints, as demonstrated in Sreenath et al. (2024) and Okazaki et al. (2021). Despite these advantages, interpretability and transparency remain primary obstacles for ML-based GMMs (Alidadi and Pezeshk, 2025), limiting their adoption in risk assessment.

Bridging parametric and nonparametric GMMs, another class of hybrid GMMs has been recently developed using symbolic learning (Chen et al., 2024; Liu et al., 2025). Rather than predefining a functional form, these models employ symbolic operations within a neural network for data-driven equation discovery. The core mechanism uses symbolic activation layers combined with L_0 regularization to prune redundant weights and promote sparsity after fully connected transformations, yielding inherently interpretable models with an explicit equation as the final output (Chen et al., 2024; Liu et al., 2025).

Miller (2019) defines interpretability as "*the degree to which a human can understand the cause of a decision.*" Complementing Miller (2019), Lipton (2018) describes transparency via simulatability (the ability to contemplate the entire model), decomposability (breaking it down into intuitive parts), and algorithmic transparency (visibility into the learning process). Post-hoc explanations address these issues by using surrogate methods to interpret developed models. Garson's (1991) global approach analyzes neural network weights to compute relative input feature importance, as applied in studies like Vemula et al. (2021) and Dhanya et al. (2018). Local methods, which focus on individual predictions, include SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017), which assigns importance scores via game theory by evaluating marginal feature contributions across coalitions (e.g., Somala et al., 2024; Fayaz et al., 2024), and Local Interpretable Model-Agnostic Explanations (LIME) (Ribeiro et al., 2016), which approximates local behavior by fitting simple models to perturbed samples (e.g., Gharagozlou et

al., 2025). However, Lipton (2018) cautions that such post-hoc interpretability may not fully reveal model behavior on unseen real-world data and can potentially mislead.

ML-based models are often viewed as "black boxes," as opposed to being interpretable. Even with strong metrics such as low mean squared error (MSE) or high coefficient of determination (R^2), this opacity hinders adoption, as it offers an incomplete view of model behavior (Molnar, 2022). Rudin (2019) argues that ML models are not inherently opaque; rather, these issues arise from poor network architecture choices, and thoughtful designs can yield inherently interpretable models, which allow humans to understand without post-hoc methods, with comparable performance. For example, using fully connected layers worsens opacity through dense and entangled connections, which obscure individual neuron contributions and impede modular decomposition (Rudin, 2019). Linear and generalized linear models, decision trees, generalized additive models (GAMs) — which model the target as a sum of smooth functions of individual predictors (Hastie and Tibshirani, 1986; Agarwal et al., 2020), and modular neural networks are some inherently interpretable networks. XGBoost, random forest, and fully connected feedforward neural networks are not inherently interpretable due to the ensemble of trees or entanglement of weights, which obscures the contribution of individual inputs. To date, the symbolic approach represents a key development in inherently interpretable GMMs by yielding explicit equations, while fully data-driven ML-based approaches remain lacking in inherent interpretability.

The Gutenberg-Richter power-law implies fewer high-magnitude events, while observational biases result in scarcer near-field records. Conventionally, GMMs are trained by optimizing log-likelihood, often via MSE loss under a Gaussian assumption, which leads to overfitting on abundant small-magnitude, far-field data and underfitting on sparse, large-magnitude near-field records. Si and Midorikawa (2000) and Morikawa and Fujiwara (2013) considered weights based on distance, obtained through trial and error, with higher weights assigned to near-field records than far-field ones. This imbalance particularly affects fully data-driven ML-based GMMs with limited seismological constraints on the input scaling and attenuation patterns. To overcome the underfitting problem due to imbalanced data in such cases, Kubo et al. (2020) applied weighting to records based on peak ground acceleration (PGA) values via trial and error. However, this resulted in overfitting; consequently, they used a traditional GMM as a base model and developed an ML model on the residuals (differences between targets and base model predictions) without weighting. Kuehn (2023) recently developed a robust Bayesian regression model with Student's t-distribution for residuals to accommodate outliers and non-Gaussian variability. In the frequentist approach, no systematic methods have been proposed to overcome data imbalance other than trial-and-error weighting of records.

To bridge these gaps in systematic data handling and inherent interpretability, this study introduces two key innovations.

- a. novel HazBinLoss function that assigns a weight to each record by combining PGA-based hazard importance with inverse weighting according to the number of records in its corresponding magnitude-distance bin.
- b. A linearly additive GAM-based "glass-box" GMM (contrasting "black-box") for PGA and PSA predictions is developed using global shallow crustal data in a frequentist approach. Our proposed approach ensures additivity and decomposability throughout the network architecture by avoiding interacting pathways in all layers, achieving an end-to-end fully interpretable, fully data-driven model with monotonic constraints.

2 Strong Motion Database

The present study develops a GMM using the global shallow crustal data from various strong motion databases for the 5% damping to PGA, peak ground velocity (PGV), and response spectrum (PSA). The NGA-West2 (Ancheta et al., 2014) contains data corresponding to regions from Western North America (e.g., California), China, Mediterranean (Italy, Greece, and Turkey), Japan, New Zealand, Taiwan, and other regions; however, California, Japan, and Taiwan regions dominate the database accounting for nearly 71%, 9.1%, and 9.3%, respectively. To obtain a better representation of remaining regions, data from corresponding regional flatfiles for Turkey (Sandıkkaya et al., 2024), New Zealand (Hutchinson et al., 2024), and Italy (Lanzano et al., 2022) are considered instead of those from the NGA-West2 database. The following criteria are applied to screen the data and obtain the final data for developing the model:

1. Consider events corresponding to shallow crustal tectonics with $Z_{\text{TOR}} \leq 20$ km.

2. Remove records with no metadata for inputs (magnitude, distance, V_{s30} , Z_{TOR}) and targets.
3. For New Zealand data, consider strong motion data with channels corresponding to BN and HN.
4. Only consider records with rupture distance (R_{rup}) ≤ 300 km and $M_w > 3$.
5. Remove records with sensor depth > 2 m and the longest usable periods < 5 s.
6. Remove problematic records, e.g., late P-trigger, spectral quality, pulse type records.
7. Finally, consider events with at least 5 records for computing reliable mixed effects.

The final selection contains 20,998 records from California, China, Italy, Japan, New Zealand, Taiwan, and Turkey: 3900, 228, 1037, 1723, 3820, 1485, and 8805 records, respectively. A total of 962 earthquake events recorded at 3970 stations are present in the data that was screened. Fig. 1 and Table 1 illustrate the ranges of magnitude, rupture distance, and other input features for the screened global data. Magnitudes span from 3.3 to 7.9, with California, New Zealand, and Turkey nearly covering the full range, though sparsely for $M_w > 7$. In contrast, China, Italy, Japan, and Taiwan primarily include higher magnitudes. Rupture distances extend from very close to the fault (≈ 0.02 km) to far-field up to 300 km across most regions, except Italy and Taiwan, which have limited far-field records; notably, fewer than 10 records are very close to the fault with $R_{rup} < 0.6$ km, so these are not shown in Fig. 1. All regions encompass soft to hard rock sites (V_{s30} from ≈ 50 m/s to 1749 m/s), with Japan featuring very soft sites near 50 m/s. Z_{TOR} values range from surface ruptures (0 km) to a maximum of 20 km, though Japan includes only shallow ruptures and Taiwan lacks deep ones.

Table 1. Input parameter ranges for the considered global crustal database

Region	Region Flag	M_w		R_{rup} (km)		V_{s30} (m/s)		Z_{TOR} (km)	
		Min	Max	Min	Max	Min	Max	Min	Max
California	1	3.34	7.28	2.5	300	116	1464	0	20
China	2	5.5	7.9	0.05	296.2	227	648	0	14.3
Italy	3	5.5	6.9	0.03	200	200	1286	0	18.5
Japan	4	6.6	6.9	0.02	300	49	1432	0.2	4
New Zealand	5	3.3	7.8	0.7	300	120	1500	0	20
Taiwan	6	5.9	7.6	0.6	172	124	1525	0	12
Turkey	7	3.5	7.8	0.1	300	131	1749	0	19.7

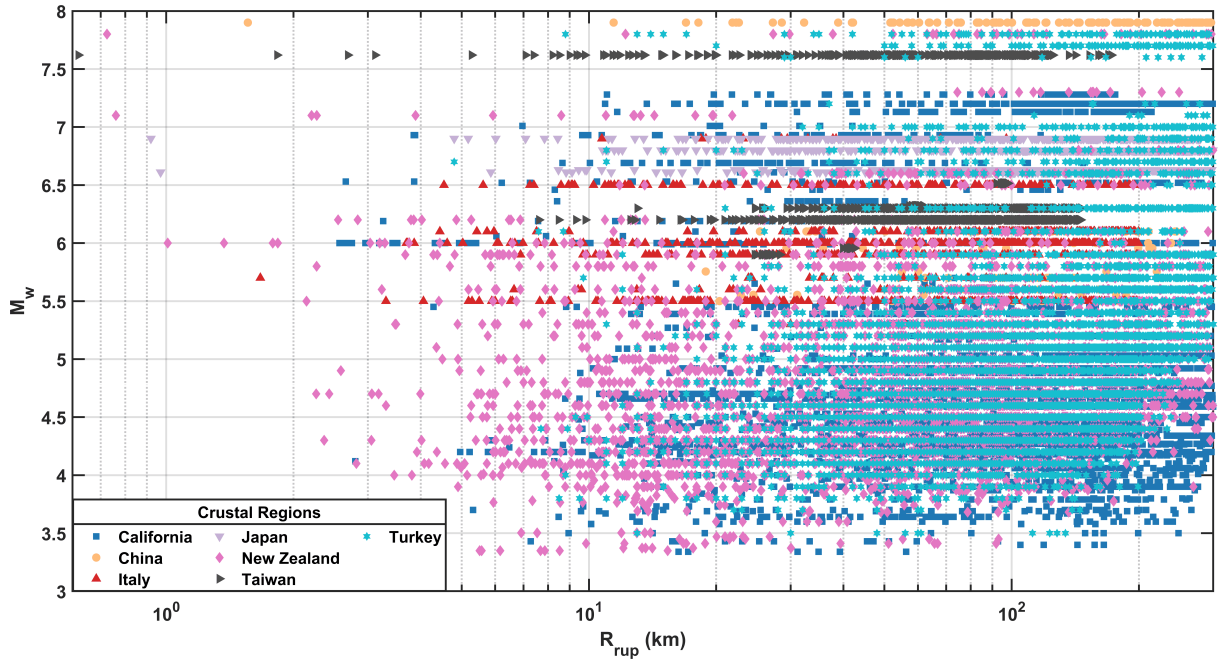


Figure 1: Magnitude and rupture distance plot for different regions in the screened data.

These uneven ranges across regions imply that predictions for some areas may involve extrapolation (e.g., high magnitudes or deep Z_{TOR} in underrepresented regions), increasing underfitting risks, while interpolation in well-covered areas could lead to overfitting. Critically, the dataset has limited large-magnitude near-field records compared to the abundance of lower-magnitude far-field data, potentially causing the model to underfit in the most damaging scenarios and overfit to the less damaging scenarios. These data characteristics reveal inherent imbalances that pose challenges for model training.

Fault classification is decided based on the rake angle (λ), with events having $-150 < \lambda < -30$ corresponding to normal (fault flag = 0), $30 < \lambda < 150$ corresponding to reverse (fault flag = 1), and other scenarios corresponding to strike-slip (fault flag = 2) classification. Records that did not have sedimentary basin depth values (Z_1 , depth to the shear-wave velocity of 1.0 km/s) are computed using the equation provided by Chiou and Youngs (2014). Additionally, California, China, Italy, Japan, New Zealand, Taiwan, and Turkey are assigned region flags ranging from 1 to 7, respectively, as shown in Table 1.

Fig. S1 plots histograms of several key input variables. The magnitude distribution is right-skewed as discussed previously, with a peak number of events between 4.2 and 4.6. A considerable decline in available events between 7 and 8 is evident. The distance histogram is left-skewed, showing a sharp decline in record counts near the fault compared to the far-field, with an order-of-magnitude difference (e.g., thousands at intermediate distances vs. hundreds below 10 km). The data also include a slightly lower number of very-soft soil records with $V_{s30} < 100$ m/s and deep basin records with $Z_1 > 1000$ m. Similarly, fewer deeper rupture events ($Z_{\text{TOR}} > 15$ km) are available in comparison to surface rupture events. Critically, limited large-magnitude events very near the rupture are present, as only a handful of records yield negative $M_w \times \ln(R_{\text{rup}})$ values.

We model magnitude scaling, style of faulting, magnitude saturation, anelastic attenuation, geometric spreading, depth scaling, basin depth effects, linear site amplification, and regional effects. Hanging-wall effects, directivity effects, and aftershock scaling are not modelled owing to the paucity of input data. Nonlinear site amplification is not modelled as one needs predicted PGA as input, which necessitates developing another model. The outputs of the model are the PGA and PSA at 25 periods (0.01, 0.02, 0.03, 0.04, 0.05, 0.075, 0.1, 0.12, 0.15, 0.17, 0.2, 0.25, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.25, 1.5, 2, 3, 4, and 5 s). The following section discusses the development of the novel HazBinLoss.

3 Implementation of HazBinLoss

In this section, details of the proposed loss are described. Kubo et al. (2020) initially considered weighting records based on PGA to address data imbalance, but their final model used a traditional parametric GMM as a base. Building upon their work, this study develops a novel loss termed HazBinLoss (Eq. 1). By applying weights at the mini-batch level, HazBinLoss mitigates underfitting in sparse, large-magnitude near-field regions, thereby enhancing model generalization. Note that a single weight is assigned to each record, consistent across all periods. HazBinLoss comprises two complementary terms: a global hazard-informed component and an adaptive bin count component. The hazard-informed component based on PGA values is computed once for the entire training set, while the bin count component is computed adaptively for each mini-batch based on the number of records in each bin. Here, a bin refers to a 2D map defined by magnitude and distance. The M_w bins are divided into ranges: 3.0–4.0, 4.0–5.0, 5.0–6.0, 6.0–7.0, 7.0–7.2, 7.2–7.4, 7.4–7.6, 7.6–7.8, and 7.8–8.0. The R_{rup} bins are grouped as 0–20 km, 20–50 km, 50–100 km, and 100–300 km. These components are combined using a hyperparameter (α), given in Eq. 2. Finally, a sigmoid scaling is applied to obtain $W_{b(k)}$ to ensure the weights vary smoothly using Eq. 3 and its parameters chosen to avoid saturation. The combined weights, as visualized in Fig. 2, combine bin count and hazard importance to prioritize high-damaging large-magnitude near-field regions through higher values in the top-left areas, while de-emphasizing abundant low-magnitude far-field bins, elsewhere.

$$\mathcal{L}_{\text{HazBin}} = \frac{1}{N} \sum_{k=1}^N W_{b(k)} \cdot (\hat{y}_k - y_k)^2 + \lambda \|\theta\|_2^2 \quad (1)$$

In Eq. 1, N is the mini-batch size, $W_{b(k)}$ is the combined weight (size = $N \times 1$) assigned to the bin containing sample k , $b(k)$ denotes the bin index function that maps sample k to its corresponding 2D bin

based on M_w and R_{rup} values, y_k and $\hat{y}_k$ are target and predicted values for sample k (size = $N \times 27$, where 27 corresponds to PGA and the number of PSA periods), λ is the L_2 regularization coefficient, and θ represents the model parameters. The underlying MSE operates on $N \times 27$ tensors, with the weight broadcast across columns.

$$W_{b(k)}^{raw} = (1 - \alpha)B_k + \alpha H_k \quad (2)$$

In Eq. 2, α is a tunable hyperparameter ranging from $[0, 1]$, with α values close to zero implying higher weighting to data distribution based on bin counts (B_k) and α values close to 1 implying higher weighting to hazard importance (H_k).

$$W_{b(k)} = \frac{1}{1 + \exp\left(-4\left(W_{b(k)}^{raw} - 0.5\right)\right)} \quad (3)$$

3.1 Hazard-Informed Component

The hazard-informed component draws inspiration from physics-informed neural networks. Similar to Kubo et al. (2020), who weighted records based on PGA values to emphasize potential hazard, this component assigns higher values to high-magnitude near-field records, which are critical for hazard assessment but underrepresented due to the power-law distribution discussed in the introduction. Lower values are assigned to low-magnitude far-field records. To compute this, we fit a simple GMM for PGA using magnitude and distance terms with a functional form similar to Idriss (2014), given by Eq. 4. This equation is evaluated at bin mid values to obtain PGA-based importance values and are normalized by its maximum value to obtain H_k , ranging from $(0, 1]$. Intuitively, higher PGA values (e.g., $M_w > 7$ and $R_{rup} < 20$ km) imply a higher hazard-informed component, as they receive elevated weights as demonstrated in Fig. 2, subplot (b).

$$\ln(PGA) = c_0 (M_w - c_1) + c_2 (M_w - c_1)^2 + (c_3 + c_4 M_w) \ln\left(\sqrt{R_{rup}^2 + c_5^2}\right) + c_6 R_{rup} \quad (4)$$

In Eq. 4, c_0 to c_6 are the coefficients determined by the nonlinear regression. $c_0 = 0.8959$, $c_1 = 3.814$, $c_2 = -0.17$, $c_3 = 2.683$, $c_4 = -0.263$, $c_5 = 5.046$, and $c_6 = -0.006435$ (MSE: 0.7307, MAE: 0.6777, R^2 : 0.8332). This equation considers first-order and second-order magnitude scaling, geometric attenuation, anelastic spreading, and magnitude-distance dependent attenuation terms. To reduce complexity, this functional form is region-agnostic and independent of V_{s30} , Z_{TOR} , and other effects, avoiding additional nested loops per mini-batch.

3.2 Bin Count Component

The bin count term addresses the data imbalance issue by weighing inversely based on the sample density in each bin. It ensures that abundant low-magnitude far-field records get lower weight, and sparse high-magnitude near-field records get higher weight. For each bin (i, j) , the count weight $C_{i, j}$ is computed by Eq. 5. After trial and error, the logarithmic form (inspired by inverse square root proportionality) is chosen as it smoothly scales weights, with denser bins receiving lower values (but not too aggressively low) to de-emphasize them. The values obtained from Eq. 5 are normalized by dividing by the maximum $C_{i, j}$ value to obtain B_k values, which range from $(0, 1]$. Fig. 2, subplot (a), illustrates this clearly for the bin count term, conveying the above description.

$$C_{i, j} = 1 + \ln\left(\frac{\max(count_k)}{\max(count_k, 5)}\right) \quad (5)$$

In Eq. 5, $count_k$ is the number of records in each 2D magnitude-distance bin, $\max(count_k)$ is the maximum count across bins in the mini-batch (making it batch-adaptive), and a minimum of 5 ensures near-empty

bins are not excessively weighted. In the following section, details of the developed architecture are discussed.

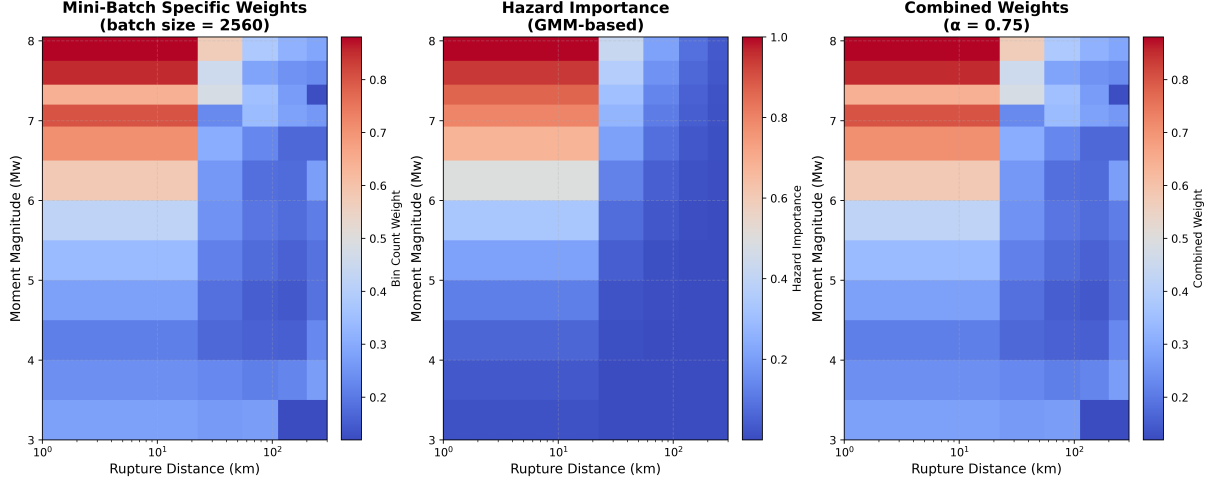


Figure 2. The plot illustrates mini-batch weights, hazard importance weights, and finally combined weights for a particular sample.

4 Details of Interpretable Architecture

In this section, we implement an inherently interpretable fully data-driven neural architecture. Traditional neural networks can learn complex nonlinear patterns on seismic data because input features are allowed to entangle and have unrestricted interactions. However, this obscures individual feature contribution to the final output, leading to challenges such as decomposability and simulatability, as defined by Lipton (2018), compromising transparency and hindering adoption in hazard analysis. We address these critical issues by developing a GAM-inspired neural architecture that processes input features additively through non-interacting pathways (Hastie and Tibshirani, 1986; Agarwal et al., 2020), ensuring end-to-end interpretability while simultaneously taking advantage of data-driven flexibility. This design allows for linear additivity across all the input features without the need for post-hoc explanations, aligning with Rudin (2019). In simple terms, each pathway corresponds to a neural network with a single input, processing features independently before additive summation to form the final prediction, given by Eq. 6. This network architecture is illustrated in Fig. 3.

$$\begin{aligned} \ln(\text{Target}) = & b + f_{\text{mag}}(M_w) + f_{\text{mag-FM}}(M_w, FM) + f_{\text{geo}}(\ln R_{\text{rup}}) + f_{\text{anelas}}(R_{\text{rup}}) \\ & + f_{\text{insite}}(\ln V_{s30}) + f_{\text{depth}}(Z_{\text{TOR}}) + f_{\text{basin}}(\ln Z_1) \\ & + f_{\text{mag-geo}}(M_w \ln R_{\text{rup}}) + f_{\text{reg}}(\text{Region flag}) + \text{Residue} \end{aligned} \quad (6)$$

In Eq. 6, Target denotes the vector consisting of [PGA, PGV, PSA], b is a learnable bias, and each $f(\cdot)$ is a dedicated sub-network for each effect we model and learned through backpropagation. Residue is the remaining residual or irreducible noise corresponding to aleatory variability, which cannot be explained with additional inputs or more data. Additionally, monotonic constraints are enforced post-batch during training to ensure physical consistency on magnitude, geometric and anelastic distance, linear site, and depth layers. The implementation details are provided in Sreenath et al. (2024) and Okazaki et al. (2021) and thus are not provided here to avoid repetition. Therefore, the proposed architecture captures: magnitude scaling using M_w term, style of faulting effects through M_w and FM flag terms, magnitude saturation through $M_w \times \ln(R_{\text{rup}})$ term, anelastic attenuation through R_{rup} term, geometric spreading using $\ln(R_{\text{rup}})$ term, depth scaling using Z_{TOR} , basin depth effects using $\ln(Z_1)$, linear site amplification using $\ln(V_{s30})$, and regional effects using regional flag term. It should be remembered that the first hidden layer output from the $f_{\text{mag}}(M_w)$ is passed as an additional input to the second hidden layer of $f_{\text{mag-FM}}(M_w, FM)$ to enable controlled interaction, while still maintaining interpretability. NGA-West2 models consider FM dependent on M_w . Hence, these two inputs are considered for modeling the style of faulting term.

To conclude, by isolating effects into additive components, the network achieves decomposability. The subsequent section discusses the performance of the proposed architecture.

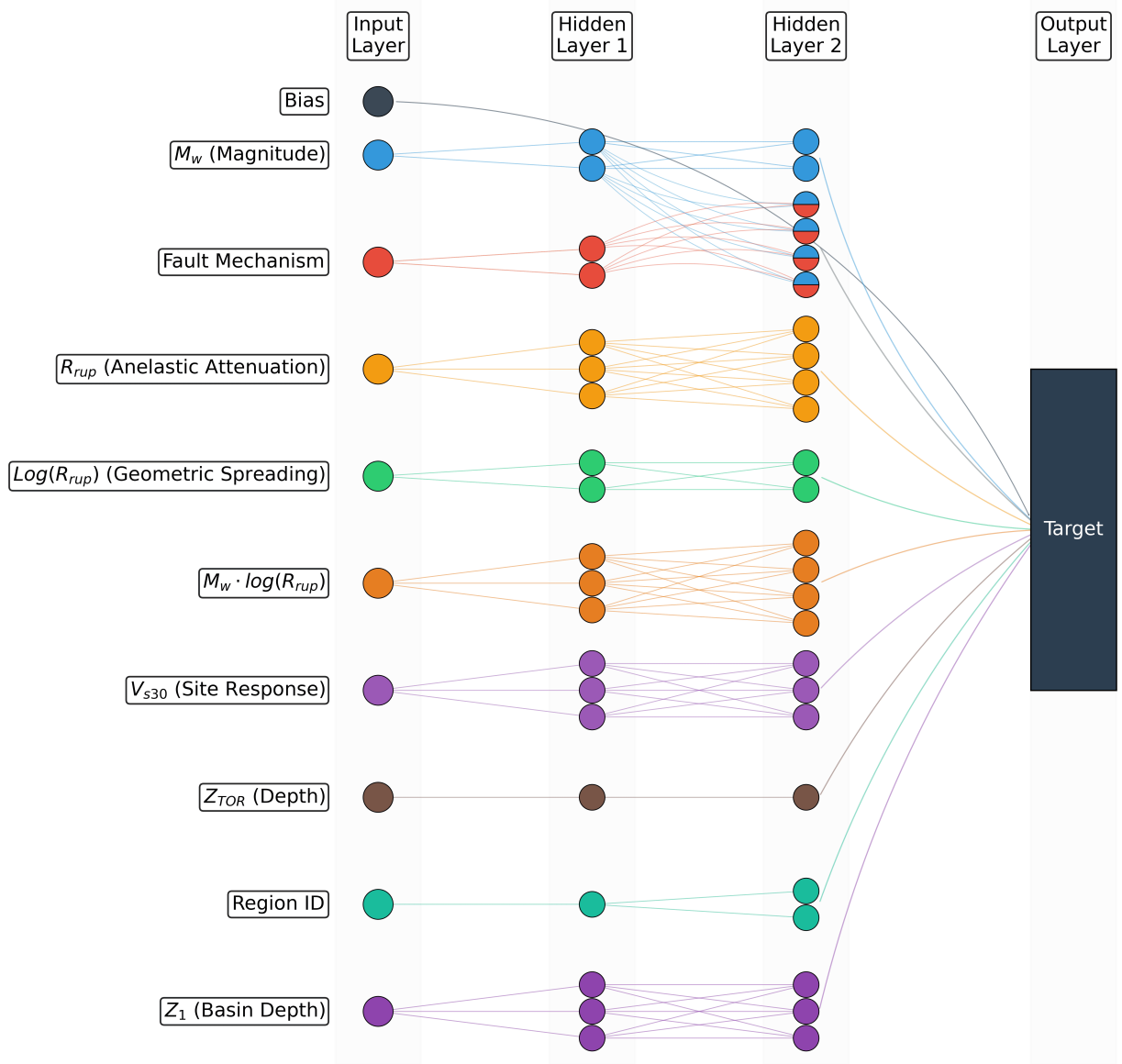


Figure 3. Interpretable network architecture with independent pathways for each input.

The earthquake events from the screened data are divided into 70%, 15%, and 15% for training, validation, and testing sets, so that no records from the same event would be split across sets, thereby preventing intra-event correlation leakage and ensuring independent evaluation. The model is trained in the PyTorch framework (Paszke et al., 2019) with Optuna hyperparameter optimizer (Akiba et al., 2019). Additionally, the work also makes use of the following Python libraries: Pandas (McKinney, 2010), Numpy (Harris et al., 2020), and Matplotlib (Hunter, 2007), and the R library: lme4 (Bates et al., 2015) for mixed effects. Model training was conducted on an NVIDIA RTX A1000 6GB GPU using CUDA 12.4, with each hyperparameter optimization run requiring approximately 3 minutes, depending on learning rate and batch size.

5 Results and discussion

In this section, the performance of the proposed architecture and loss function are evaluated. The model is initially trained with Optuna hyperparameter tuning using the MSE loss and HazBinLoss ($\alpha = 0, 0.25, 0.5, 0.75, \text{ and } 1$). The MSE, mean absolute error (MAE), and R^2 metrics are then computed for each model using the test records, as shown in Table 2. It can be observed that the model $\alpha = 0.25$ achieves

the best performance on all the test records, and the remaining models have comparable values. When the metrics are computed using all the records corresponding to $M_w \geq 7$ and $R_{rup} \leq 50$ km, the MSE model significantly underpredicts high damage causing records compared to the models developed using any of the α values in HazBinLoss. It implies that the models developed using the HazBinLoss do not result in lower overall errors but do result in lower errors for large-magnitude near-field records.

Furthermore, when evaluating the most critical records (M_w 7.8–8.0 and R_{rup} 0–20 km), the MSE loss model shows poor performance ($MSE = 1.3215$) compared to the optimal HazBinLoss model with $\alpha = 1$ ($MSE = 0.5150$). Since the MSE loss model has similar test set performance with HazBinLoss models but poor performance for $M_w \geq 7$ and $R_{rup} \leq 50$ km records, which have limited recorded data, it might have slightly overfit bins with abundant data. Additionally, since near-field ground motions are dominated by source and path effects, and less significantly by the site effects, the $\alpha=1$ model achieves superior performance in these critical scenarios. In the far field, all the models have similar metrics, with the MSE loss model achieving marginally better performance, perhaps due to random initialization or slightly overfitting. We conclude that the underweight assigned did not negatively influence the HazBinLoss models to behave poorly at bins with ample records.

Table 2. Model performance using MSE loss and HazBinLoss for different α values

Model (Loss)	Test-set metrics			$M_w \geq 7, R_{rup} \leq 50$ km		High-mag bins	
	MSE	MAE	R^2	MSE	MAE	$M_w \geq 7.8, R_{rup} \leq 20$ km	$M_w \geq 7.8, R_{rup} \geq 100$ km
MSE Loss	0.5932	0.6038	88.59	0.6752	0.6657	1.3215	0.5515
$\alpha = 0$	0.6069	0.6172	86.20	0.4890	0.5626	0.6674	0.5871
$\alpha = \mathbf{0.25}$	0.5402	0.5795	88.51	0.4654	0.5444	0.6798	0.5896
$\alpha = 0.5$	0.5867	0.6068	88.13	0.4464	0.5394	0.5993	0.5862
$\alpha = 0.75$	0.5942	0.6072	86.50	0.4608	0.5469	0.5717	0.6110
$\alpha = 1$	0.5664	0.5982	88.66	0.4025	0.5098	0.5150	0.5987
Models retrained, removing $M_w \geq 6$ and $R_{rup} \leq 100$ km data from training and validation sets.							
MSE Loss	0.6006	0.6128	90.35	0.7086	0.6747	1.0761	0.5891
$\alpha = 0$	0.5958	0.6111	88.04	0.6829	0.6621	0.9322	0.6520
$\alpha = 0.25$	0.6298	0.6281	89.03	0.6123	0.6174	0.5927	0.5726
$\alpha = 0.5$	0.6160	0.6229	89.41	0.6191	0.6209	0.8615	0.5754
$\alpha = 0.75$	0.5959	0.6112	89.05	0.5828	0.6019	0.5345	0.5564
$\alpha = 1$	0.6128	0.6203	89.13	0.5778	0.6009	0.5646	0.5636

Ablation study: To further illustrate the efficacy of HazBinLoss, all the records corresponding to $M_w \geq 6$ and $R_{rup} \leq 100$ km are removed from the training and validation sets. The models are retrained without updating the Hazard term in Eq. (4), and the results are shown in Table 2. The overall MSE and MAE metrics, when computed for the test set for all the models, are close, with a slight increase in error metrics compared to the previous case due to data removal. Additionally, the MSE loss model performed poorly, similar to the earlier case. Notably, the $\alpha = 0$ model also performs poorly because bin count weighting becomes ineffective without actual high-magnitude near-field records to learn from. For other α values, the hazard importance term contributed to reducing the loss, which demonstrates that the hazard-informed term enables learning about critical scenarios even when direct training data is limited.

Model comparison: A traditional data-driven feedforward network is trained with HazBinLoss ($\alpha = 0.73$), incorporating inputs: M_w , R_{rup} , $\ln(R_{rup})$, $\ln(V_{s30})$, $\ln(Z_1)$, Z_{TOR} , FM, and Region Flag (excluding the $M_w \times \ln(R_{rup})$ interaction term, as the weight interaction captures this). The test set performance metrics for this model are: $MSE = 0.5612$, $MAE = 0.5954$, and $R^2 = 89.13\%$. These results compare favorably with the interpretable model ($\alpha = 0.25$) metrics presented in Table 2, with the interpretable model achieving marginally superior accuracy, likely attributable to differences in optimization convergence and random weight initialization. For reference, the performance metrics of established NGA-West2 models on the screened dataset show: BSSA14 with $MSE = 0.95$ and $MAE = 0.77$, and CB14 with $MSE = 0.88$ and $MAE = 0.73$. Notably, the database contains a substantial proportion of events from regions outside the

original NGA-West2 dataset, which likely contributes to the elevated prediction errors observed in these models.

It should be noted that since the model with $\alpha = 0.25$ achieved better performance on entire test set and not to one bin combination, it is considered for the subsequent analysis. Though, a better performance metric could be designed to weigh between global vs critical bins performance, this is beyond the scope of this work. The analysis demonstrates that interpretable model architectures with independent additive pathways achieve comparable predictive accuracy relative to the fully data-driven approach and substantial performance improvements over both NGA reference models.

Ground Motion Model Component Effects

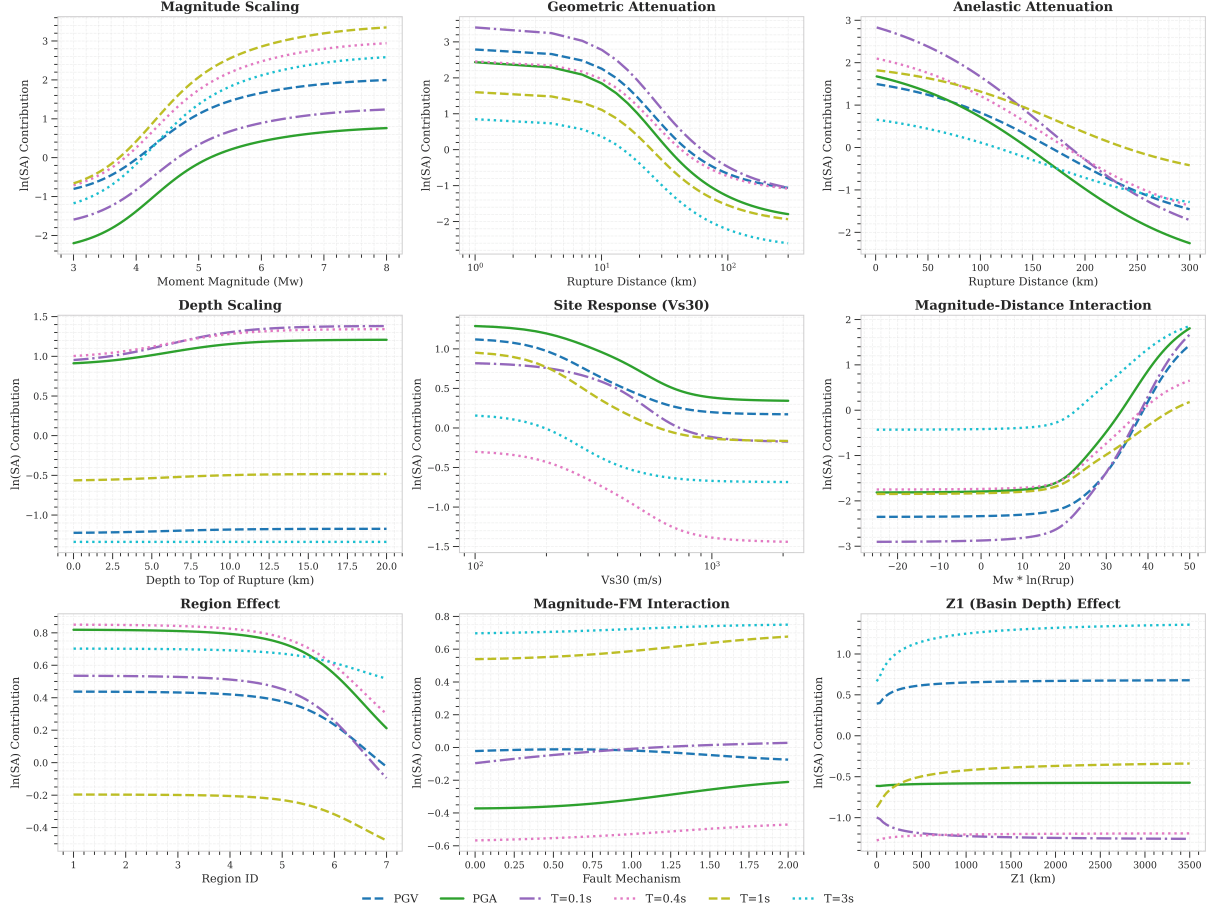


Figure 4. Pathway scaling for the interpretable model with $\alpha = 0.25$

5.1 Interpretable Model Scaling

This section analyzes the pathway scaling of the model with $\alpha = 0.25$, as illustrated in Fig. 4. All pathways exhibit period-dependent behavior consistent with established seismological principles.

Source effects: The magnitude pathway term initially grows rapidly for smaller magnitudes before saturating (for near-field distances) or transitioning to linear growth (far-field distances), mirroring the magnitude scaling in BSSA14, which exhibits parabolic growth up to a hinge magnitude followed by saturation or linear behavior. The magnitude-fault mechanism interaction exhibits linear scaling, with strike-slip faulting producing the highest values, followed by reverse and normal faulting, a pattern consistent with CB14. The depth scaling term (Z_{TOR}) increases up to 10 km before saturating at lower periods, while remaining relatively constant at intermediate and higher periods, differing from NGA-West2 models, which assumed linear dependence across all periods.

Path Effects: The anelastic attenuation term exhibits linear dependence on distance, similar to the region-dependent attenuation in BSSA14. Geometric attenuation shows approximately linear behavior between 10-100 km with near-fault saturation effects similar to BSSA14's functional form, and steeper

attenuation beyond 100 km. While NGA-West2 models employ the $\sqrt{R^2 + h^2}$ term (where R is the distance parameter and h is a fictitious distance parameter) to capture near-fault saturation, the $\ln(R_{rup})$ term inherently captures this effect. Similar conclusions are observed from Fig. S2, where distance attenuation for different magnitude combinations is illustrated.

Magnitude-Distance Effects: The magnitude-distance interaction term exhibits saturation at lower values (corresponding to large-magnitude near-field scenarios where saturation is expected) and linear growth at higher $M_w \times \ln(R_{rup})$ values, consistent with BSSA14's $M_w \times \ln(\sqrt{R^2 + h^2})$ formulation. The model also captures other well-known phenomena. Lower periods at higher magnitudes and near-field distances show converged predictions, as evident in Fig. S2. Additionally, lower periods show pronounced attenuation with the increase in distance. Finally, the gap between different magnitude levels reduces as distance increases, with a pronounced effect at the lower periods compared to the higher periods.

Site effects: The site response term shows a nonlinear decrease from very-soft to rock sites before saturating at hard-rock conditions. Higher values for very-soft sites imply significant spectral amplification compared to other site conditions. Unlike the BSSA14 linear dependence assumption, our model captures the expected nonlinear site response, which is evident from its non-uniform slope. Additionally, period-dependent site amplification is evident. Basin depth effects demonstrate exponential growth at shallow depths, followed by saturation, consistent with CY14's functional form. Higher periods show significant basin amplification with limited amplification at intermediate periods and de-amplification effects at lower periods, as evident by the slope of the curves.

Region effects: Turkey exhibits the most negative regional adjustment factors, corresponding to the highest attenuation relative to other regions. Taiwan shows slightly higher values than Turkey, while California demonstrates the highest amplification. China and Italy exhibit regional amplifications similar to those in California. Additionally, regional effects are most pronounced for high-magnitude, far-field scenarios (e.g., $M_w=7.5$, $R_{rup}=300\text{km}$) and less significant for low-magnitude, near-field conditions (e.g., $M_w=3.5$, $R_{rup}=10\text{km}$), as illustrated in Fig. S3. Regional dependency exhibits strong period-dependence, with short periods showing the highest regional variations across all magnitude-distance combinations, while long periods demonstrate minimal regional differences with overlapping response spectra. The model has learned region-specific path effects that could be further enhanced by coupling the regional flag with the path terms.

5.2 Component Contribution vs SHAP analysis

In this subsection, the mean absolute values of pathway component contributions from the interpretable model are illustrated for PGA, PSA at 0.1s, and 1.0s in Fig. 5. The corresponding SHAP values are computed as post-hoc explanations and shown alongside for comparison. Each variable uses consistent color coding across subplots for better visualization. The pathway contribution plots (top row) for PGA, PSA 0.1s, and PSA 1.0s clearly show that distance (geometric spreading and anelastic attenuation), magnitude, and magnitude-distance terms dominate model attribution at all periods, with mean contributions often around or above 1 for these features. Bias and fault mechanism term contributions remain consistently minor. In contrast, only magnitude and anelastic attenuation terms significantly contribute to the SHAP plots, with their contribution above 1. Furthermore, only the contribution of the anelastic attenuation term is correctly predicted by the SHAP model in the PGA plot. Magnitude and site term contributions are more accurately approximated, and remaining component contributions are poorly estimated in the PSA at 0.1s. Finally, except for the magnitude, site, and anelastic attenuation terms, all the term contributions are poorly predicted for the PSA at 1s.

As discussed in the introduction, despite SHAP values having a sound mathematical basis, they are at best an approximation to the actual model. Except for a few feature contributions, the remaining features are poorly approximated and systematically underestimated for this model across the periods. Consequently, this analysis reinforces the value of inherently interpretable architectures over black-box models with post-hoc explanations.

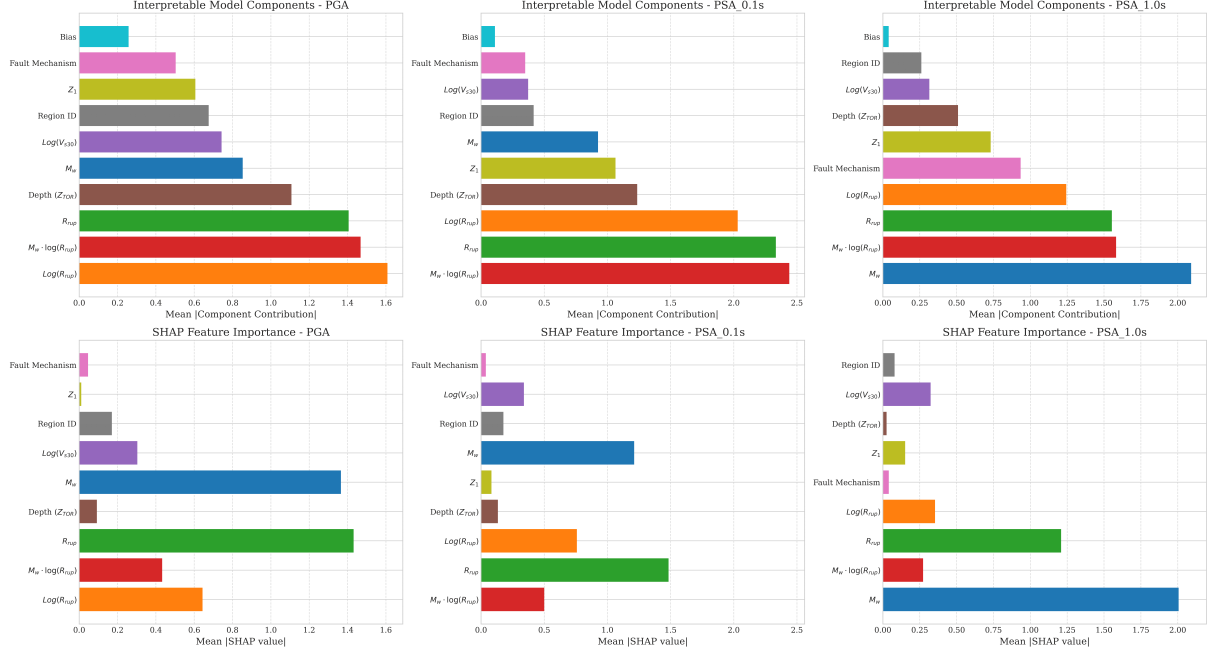


Figure 5. Mean component contribution from the interpretable pathways and SHAP analysis values

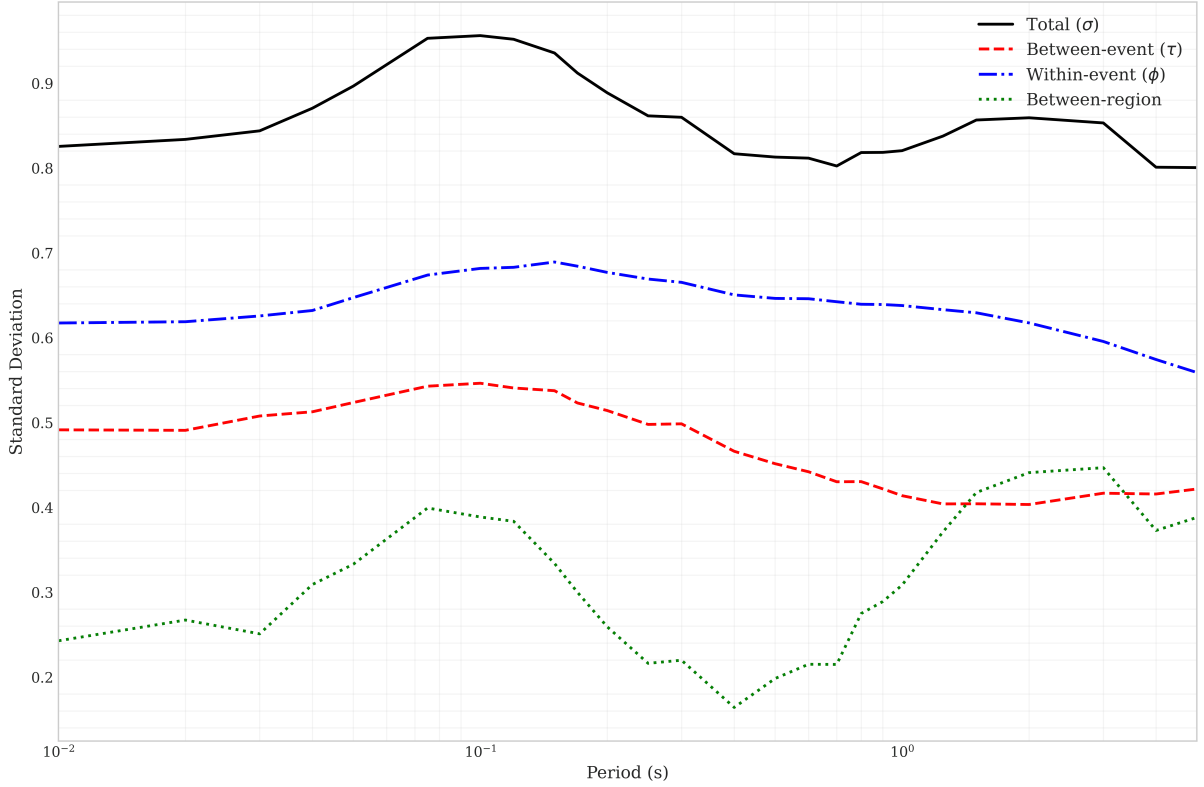


Figure 6. Mixed-effect regression analysis for the interpretable model

5.3 Mixed-Effect Analysis

Earthquake records from the same seismic event recorded at multiple stations exhibit similar properties due to common source characteristics. Additionally, recorded data originating from the same region and station have similar characteristics due to shared path and site effects. The likelihood calculation in fixed-effects models ignores this correlation structure, treating all records as independent observations. This assumption results in significant bias in variance estimates and predictions, as these records are not statistically independent. Mixed-effects regression addresses this problem by explicitly modeling

the correlation structure, producing unbiased residuals and predictions. Implementing mixed-effects for earthquake data requires nonlinear mixed-effects modeling, which must be obtained iteratively following the Abrahamson and Youngs (1992) approach. The iterative algorithm alternates between fixed-effects parameter estimation using neural networks and random-effects variance component estimation using maximum likelihood in R's lme4 package.

We considered only earthquake events with at least 5 recordings to obtain reliable random effects estimates. Of the 3,970 stations in the screened database, 1,450 stations had only one record, and merely 981 stations had at least 5 recordings. Consequently, station terms are excluded from the random effects structure. A hierarchical mixed-effects model where earthquake events are nested within regions is developed iteratively as given below (Meenakshi et al., 2023):

Iterative Nonlinear Mixed-Effects Algorithm:

- Input: $X, y^0, R, E, \epsilon, \text{max_iter}$
- Output : $\theta^*, \sigma, \tau, \phi_r, \phi$
- Initialise: $t \leftarrow 0, y^{(0)} \leftarrow y^0$
- Repeat:
 - $\theta^{(t+1)} \leftarrow \text{Train Interpretable Model } (\theta^{(t)}; X, y^{(t)}, \text{HazBinLoss})$
 - $\text{residual}^{(t+1)} \leftarrow y^0 - f_{\theta}(t+1)(X)$
 - $\{\delta\text{BR}, \delta\text{BE}, \delta\text{WER}\} \leftarrow \text{MixedEffects}(\text{residual}^{(t+1)} \sim 1 + (1|R/E))$
 - $y^{(t+1)} \leftarrow y^{(0)} - \delta\text{WER}$
 - $t \leftarrow t + 1$
- until $|\ell^{(t)} - \ell^{(t-1)}| < \epsilon$ OR $t \geq \text{max_iter}$

In the above algorithm, X and y^0 are the input and target vectors, and $y^{(t)}$ is the modified target. R and E are vectors describing region and event details, ϵ and max_iter are set to be 5% and 5. δBR , δBE , and δWER are random effects estimates for between-region, between-event, and within-region-event components, respectively. Residual: $\delta\text{BR} + \delta\text{BE} + \delta\text{WER}$. σ, τ, ϕ_r , and ϕ correspond to total sigma, between-event, between-region, and within-region-event, respectively. θ^* corresponds to final weights of the model, and ℓ^t : Log-likelihood value used for convergence assessment.

The mixed-effects regression assumes a standard Gaussian distribution for residuals with zero mean and standard deviations: τ, ϕ_r , and ϕ , where the total aleatory variability is given by Eq. 7. Diagnostic analysis reveals departures from normality (Shapiro-Wilk $p < 0.001$, Kolmogorov-Smirnov $p < 0.05$) with kurtosis ranging from 0.33-1.07 (highest at short periods, lowest at long periods). However, outlier percentages remain low (1.39-1.70%). Consequently, the Gaussian assumption is acceptable and future work could explore heavy-tailed distributions. The obtained variance components are plotted in Fig. 6, showing slightly higher sigma values than NGA-West2 models due to the significantly larger database used for training. The τ, ϕ_r, ϕ , and σ values range in Fig. 6 between 0.4033-0.5463, 0.1641-0.4467, 0.5542-0.6892, and 0.7449-0.9460, respectively.

$$\sigma = \sqrt{\tau^2 + \phi_r^2 + \phi^2} \quad (7)$$

The computed inter-event, inter-region, and intra-region-event residuals are plotted to understand whether any bias exists in the developed model with respect to input variables. Often, an underfit or overfit model exhibits some bias. An underfit model fails to capture underlying relationships, leading to systematic residual patterns across predictor ranges. Fig. 7 illustrates the inter-region and inter-event residuals plotted against magnitude and region flag, respectively, for PGA (representing low periods), PSA at 0.3s (representing intermediate periods), and PSA at 3s (representing high periods). The error bars are plotted with the mean and standard deviation obtained from the residuals within the bins considered. The mean values of inter-region residuals are plotted at zero with standard deviation equal to the inter-region residual value corresponding to each regional flag.

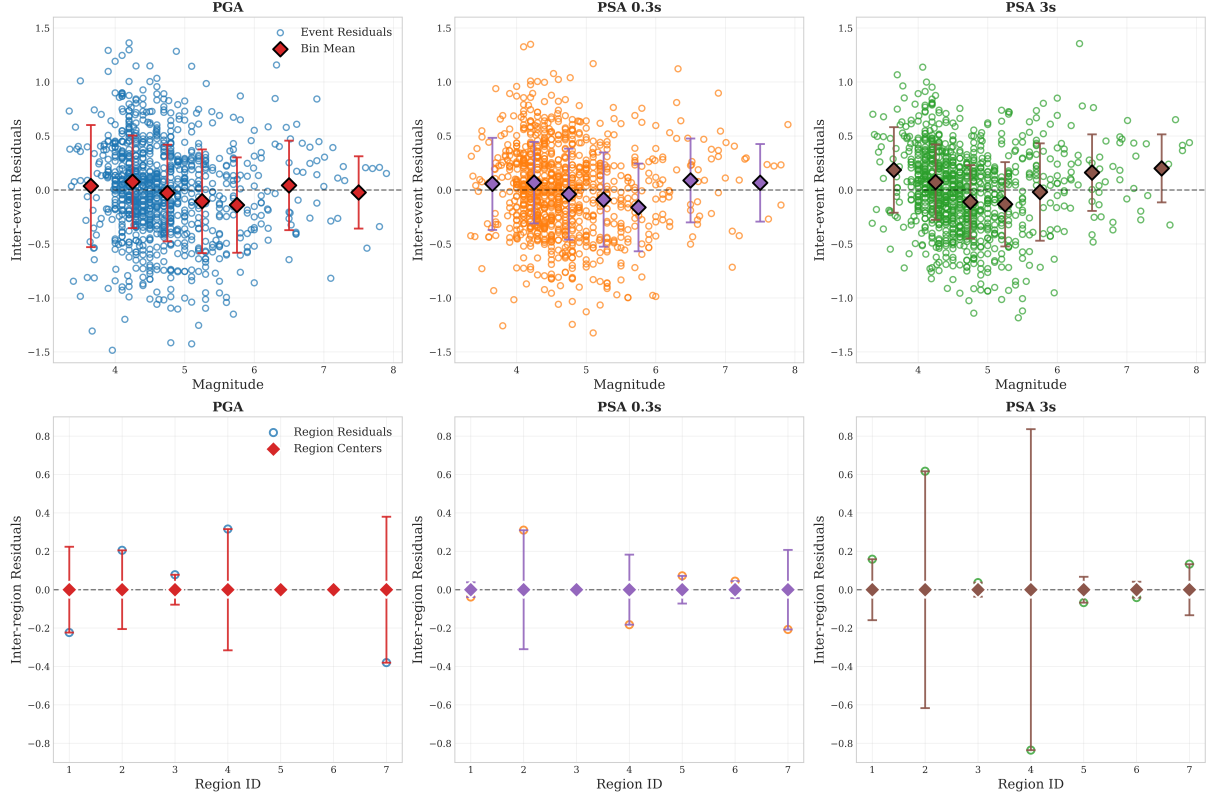


Figure 7. Inter-event and inter-region residuals plotted against magnitude and regional flags, respectively, for PGA, PSA at 0.3 and 3s.

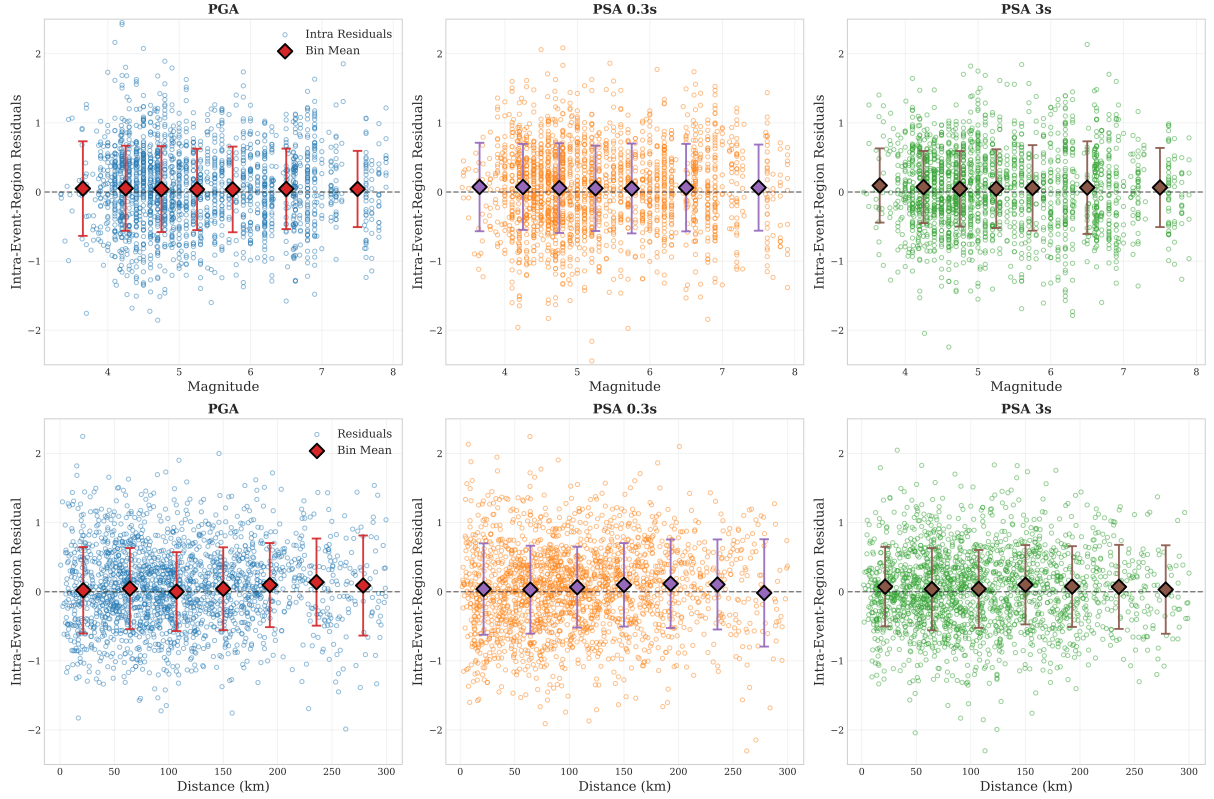


Figure 8. Intra-region-event residuals plotted against magnitude (top plots) and distance (bottom plots), respectively, for PGA, PSA at 0.3 and 3s.

The varying inter-region standard deviations reflect the heterogeneity - regions with minor standard deviations indicate more homogeneous seismic characteristics, while larger standard deviations suggest

greater geological complexity or diverse recording conditions within that region. Japan is observed consistently to exhibit greater diversity compared to the other regions. No bias is evident from the inter-event plots as the mean values are close to zero, and no trend is apparent with inter-event standard deviations varying slightly across magnitude bins. Mean values corresponding to the PSA 3s subplot have slightly nonzero values, especially where limited data is available.

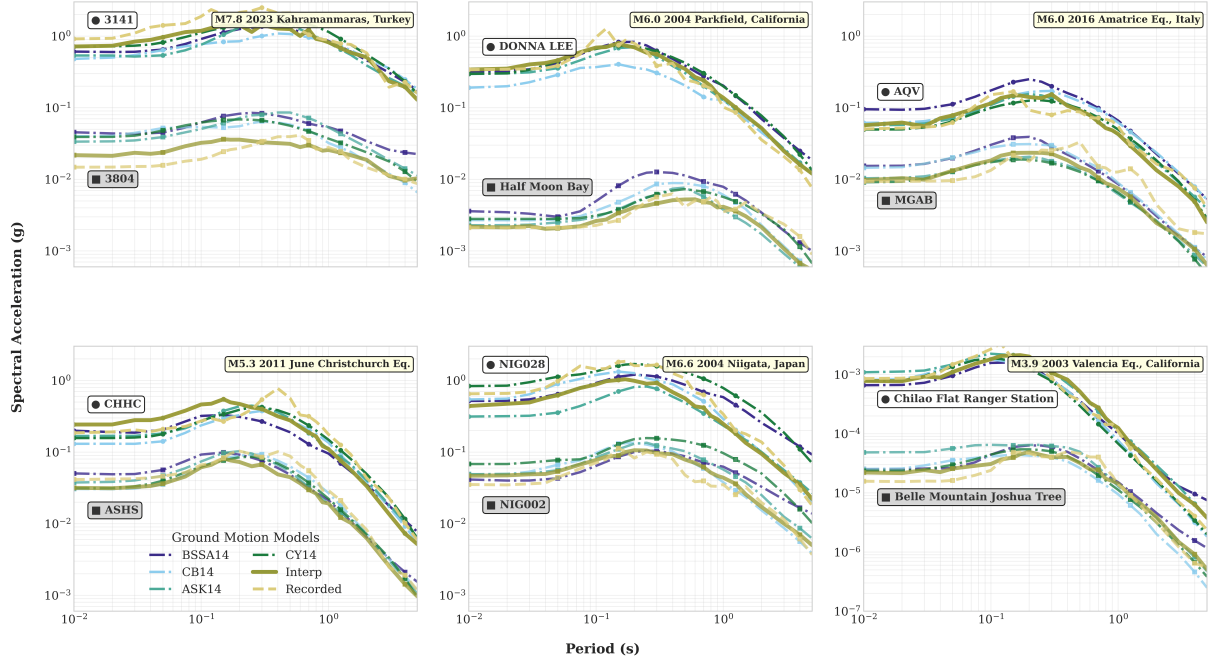


Figure 9. Comparison of response spectra predictions from the interpretable model and NGA-West2 GMMs (BSSA14, CB14, CY14, ASK14) against recorded values for six earthquakes across different regions.

Fig. 8 similarly illustrates intra-region-event residuals plotted against magnitude and distance, respectively, for PGA, PSA at 0.3s, and PSA at 3s, along with error bars. Instead of plotting all residuals, only 2000 residuals are randomly chosen and plotted for clear visualization. The mean values at all bins are very close to zero in all subplots, and the standard deviations are consistent across bins. It demonstrates that the mixed-effects modeling approach successfully removes systematic biases with respect to magnitude and distance, while HazBinLoss contributes to improved model fitting by handling the data imbalance better.

5.4 Comparison with other GMMs

The interpretable model is evaluated against established NGA-West2 GMMs across six diverse earthquake scenarios worldwide. Each subplot presents spectral predictions at two recording stations, with input parameters spanning magnitudes 3.9–7.8, rupture distances 0.1–248 km, all faulting mechanisms, depth to top of rupture 0–13.4 km, shear-wave velocities 200–930 m/s, and basin depths 10–1085 m. The comparative analysis demonstrates that the interpretable model exhibits strong agreement with both recorded spectra and NGA-West2 predictions across this comprehensive parameter space. Notably, for the Kahramanmaraş earthquake, station 3141 recorded a PGA of 0.9g, which the interpretable model predicts with exceptional accuracy. Similar robust performance is observed across the full range of spectral periods and amplitude levels. The model shows consistent performance across varying conditions, including small and large magnitudes, near- and far-field scenarios, all styles of faulting (normal, reverse, and strike-slip), surface to shallow rupture earthquakes, very soft soils to hard rock sites, and shallow to deep basin depths. While acknowledging the inherent variability in earthquake ground motion recordings, the systematic agreement between the interpretable model predictions, established GMMs, and observed data validates the model's capacity to capture spectral characteristics across diverse seismological conditions.

6 Conclusions

This study develops an inherently interpretable and transparent “glass-box” ML-based GMM for 5% damping to PGA, PGV, and PSA using global active shallow crustal data, addressing the interpretability

limitations that hinder ML adoption in seismic risk assessment. We demonstrate that ML models are not inherently "black-box"—instead, interpretability depends on the architectural choices employed during development. Our approach achieves interpretability by ensuring inputs remain disentangled at every layer through independent additive pathways. Additionally, we introduce a novel HazBinLoss function to address data imbalance in seismic datasets. This loss function combines two components via a hyperparameter (α):

- Bin count component: Applies inverse weighting based on sample density within magnitude-distance bins.
- Hazard-informed component: Inspired by physics-informed neural networks, prioritizes high-magnitude near-field records critical for hazard assessment.

Prediction results demonstrate that the component scaling of our developed model ($\alpha = 0.25$) exhibits behavior consistent with established seismological principles and period-dependent characteristics. While MSE loss significantly underpredicts critical scenarios ($M_w \geq 7$, $R_{rup} \leq 50\text{km}$), HazBinLoss substantially reduces this underprediction. Notably, the interpretable model achieves performance comparable to established NGA-West2 models and traditional feedforward networks with HazBinLoss, confirming that interpretability through independent additive pathways requires no accuracy sacrifice.

Ablation study simulates limited large-magnitude near-field data scenarios by removing all records with $M_w \geq 6$ and $R_{rup} \leq 100\text{km}$. While MSE and $\alpha = 0$ configurations perform poorly, other α values (0.25, 0.5, 0.75, 1.0) maintain significantly better performance, demonstrating that HazBinLoss—particularly the hazard-informed component—enables reliable model performance under data-limited conditions. Nonlinear mixed-effects analysis reveals no bias or trends in inter-event or intra-region-event residuals, confirming unbiased predictions. When comparing pathway component contributions against SHAP values across different periods, we find that despite SHAP's sound mathematical foundation, it remains an approximation that significantly struggles to represent component contributions accurately. Specifically, except for magnitude and shear-wave velocity contributions, all remaining parameter contributions are systematically underestimated. Finally, comprehensive validation shows that interpretable model predictions remain consistent with NGA-West2 GMMs and observed data across diverse input conditions.

This work establishes a foundation for developing data-driven seismic hazard assessment using interpretable ML architectures. Future research directions include incorporating Bayesian weights to quantify epistemic uncertainty for individual pathway components, heavy-tailed distributions for mixed-effects, fine-tuning the model for data-limited regions using HazBinLoss, coupling the regional flag with the path terms, and conducting input ablation studies to examine how component contributions vary across different scenarios. We envision that this interpretable framework will enable broader ML adoption in seismic risk assessment, providing both predictive accuracy and transparency essential for engineering decision-making.

Data Availability: All the data used to support our results and the final interpretable model will be made publicly available after the manuscript is accepted in the journal.

Author Contribution

Vemula Sreenath: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing – original draft, visualization. Filippo Gatti: methodology, resources, writing, review & editing, project administration. Pierre Jehel: resources, writing, review & editing, project administration, funding acquisition.

Declaration of generative AI in scientific writing

While preparing the manuscript, the authors used generative AI tools such as Claude 4.0 Sonnet to improve the readability and correct the language. After using these tools, the authors reviewed and edited the content as required and take full responsibility for the content of the published article.

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Declaration of competing interest: The authors declare that no competing financial interests or personal relationships could have influenced this work.

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